

Walkthrough Script — Deloitte Data Analytics Simulation

Hi, I'm Fariba. In this Deloitte data analytics simulation, I worked as an analyst for a fictional client, Daikibo Industrials, and I'll walk you through the two things I built: a machine-downtime dashboard and a pay-equality classification.

Let me start with the problem. Daikibo collected one month of sensor data from machines across four factories, and every machine sends a status message every ten minutes, either healthy or unhealthy. The client wanted to know two things: which location is losing the most time to breakdowns, and which machines are responsible.

To measure that, I created a calculated field called Unhealthy that assigns ten minutes of downtime to every unhealthy message, since that's the gap between readings. I then built two charts. The first shows total downtime by factory, and the second shows downtime by device type. I linked them so that clicking a factory filters the second chart to that location's machines.

Here's what it shows. Factory Seiko in Osaka had the most downtime by a clear margin, around 480 minutes. When I select Seiko, the device chart tells the real story: all of that downtime comes from a single machine, the Laser Welder. So this isn't lots of small failures across the floor. It's one machine with a prolonged outage. My recommendation to the client would be to prioritise maintenance and a root-cause check on that specific Laser Welder, because fixing one machine addresses the entire problem at that site.

The second deliverable looks at a different issue: pay equality. The client gave each role in each factory an equality score from minus one hundred to plus one hundred, where zero is perfectly fair. My job was to classify those scores. I added a column that labels each score as Fair if it's within ten of zero, Unfair if it's between ten and twenty out, and Highly Discriminative if it's beyond twenty in either direction. I did it with a nested IF formula so it updates automatically, and I colour-coded the results green, amber, and red so the problem areas stand out at a glance. From there, the roles flagged red, like the senior Meiyo positions, are the priority cases for a pay-band review.

So across the two tasks, the takeaway is a single point of failure driving downtime at Seiko, and a clear shortlist of roles to investigate for pay equality. Thank you.